WINDOW NEUTRALITY, FUTURE-BALANCE PROJECTION, AND THE PHANTOM BALANCE CONSTRAINT

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Abstract. We study integer-valued signals on $\mathbb{Z}$ subject to a *window neutrality constraint*: the sum over every aligned window of width W must vanish. (We fix $W = 8$ throughout, though the results hold for any $W \geq 2$.) We prove three groups of results. First, any nonzero contribution at a single tick forces a compensating balance over the remaining ticks in its window (the *future-balance theorem*), with an explicit backward projection bound $|d| \leq B \cdot R$ relating the accumulated debt d, the remaining ticks R, and the per-tick bound B. Second, we formalize the *boundary-balance rule* (the last tick of each window absorbs the residual) and prove it implies full window neutrality by a telescoping argument. Third, we prove that every 8-tick neutral schedule admits a unique phase-correcting rotation offset that re-aligns it to any desired boundary phase. These results provide the windowed-cancellation and phantom-debt tools used in companion papers on arithmetic phase bounds [2] and gauge-field defect annihilation.

1. Setup and definitions

Throughout, we fix the window width $W := 8$. (Every result below holds with W replaced by any integer ≥ 2; only the specific numerics change.)

Definition 1.1 (Window signal)**.** A *window signal* is a function $w : \{0, 1, \ldots, W-1\} \to \mathbb{Z}$. We say w is *neutral* if $\sum_{k=0}^{W-1} w(k) = 0$.

Definition 1.2 (Lock event)**.** A *lock event* at position $p \in \{0, \ldots, W-1\}$ with contribution $\delta \in \mathbb{Z} \setminus \{0\}$ is an assignment $w(p) = \delta$ for one tick of the window.

Definition 1.3 (Balance debt and remaining ticks)**.** Given a partial signal with contributions at positions $0, \ldots, k$ ($0 \leq k \leq W-1$), the *balance debt* is $d := \sum_{j=0}^{k} w(j)$ and the *remaining ticks* is $R := W-1-k$.

2. The future-balance theorem

The first result says that in a neutral window, any single tick's contribution determines an equal-and-opposite obligation on the rest.

Theorem 2.1 (Lock forces future balance)**.** *Let w be a neutral window signal. If $w(p) = \delta$ for some $p \in \{0, \ldots, W-1\}$ with $\delta \neq 0$, then*

$$\sum_{k \neq p} w(k) = -\delta.$$

Proof. By neutrality, $\sum_{k=0}^{W-1} w(k) = 0$. Split the sum into $w(p)$ and the rest: $w(p) + \sum_{k \neq p} w(k) = 0$. Hence $\sum_{k \neq p} w(k) = -w(p) = -\delta$. □

Corollary 2.2 (Nonzero contribution creates debt)**.** *After a lock at position p with value $\delta \neq 0$, the remaining $R = W-1-p$ ticks must collectively contribute $-\delta$ to achieve window neutrality.*

Date: February 2026.

2020 *Mathematics Subject Classification.* Primary 39A10; Secondary 11B83, 68Q85.

Key words and phrases. Discrete signal, window sum, neutrality constraint, balance debt, backward projection, rotational invariance.

Proof. By Theorem 2.1, the total over all other positions is $-\delta$. Positions $0, \ldots, p-1$ are already fixed (their values are determined). Hence the unfilled positions $p+1, \ldots, W-1$ (there are $R = W-1-p$ of them) must supply the remaining $-\delta$ minus whatever the earlier positions contributed. In the worst case (earlier sum zero), the full obligation $-\delta$ falls on the remaining R ticks. □

Example 2.3. With $W = 8$ and a lock $w(2) = 5$, positions $3, 4, 5, 6, 7$ (five remaining ticks) must collectively contribute -5.

3. Phantom magnitude and backward projection

We now quantify how "urgent" the outstanding debt is.

Definition 3.1 (Phantom magnitude)**.** For balance debt $d \in \mathbb{Z}$ and remaining ticks $R \geq 0$:

$$\Phi(d, R) := \frac{|d|}{R+1}.$$

(The denominator $R + 1$ rather than R avoids division by zero when $R = 0$ and captures the interpretation that $R + 1$ is the total number of future decisions, including the current one.)

Proposition 3.2 (Phantom properties)**.** (a) *$\Phi(d, R) \geq 0$ for all d, R.*

(b) *Φ is monotone increasing in $|d|$ for fixed R.*
(c) *Φ is monotone decreasing in R for fixed $d \neq 0$.*
(d) *$\Phi(0, R) = 0$: zero debt implies zero phantom.*
(e) *$\Phi(d, 0) = |d|$: when no ticks remain, the phantom equals the full outstanding debt.*

Proof. All five properties follow directly from the formula $\Phi(d, R) = |d|/(R + 1)$ and the fact that $R + 1 \geq 1 > 0$. □

Theorem 3.3 (Backward projection principle)**.** *Suppose each of the R remaining positions is bounded in magnitude by $B \geq 0$ (i.e. $|w(k)| \leq B$ for each remaining position k). Then any feasible completion satisfying window neutrality requires*

$$|d| \leq B \cdot R.$$

Equivalently: if $|d| > BR$, no feasible completion exists.

Proof. Neutrality requires $\sum_{\text{remaining}} w(k) = -d$. By the triangle inequality:

$$|d| = \Big| \sum_{\text{remaining}} w(k) \Big| \leq \sum_{\text{remaining}} |w(k)| \leq B \cdot R.$$ □

Corollary 3.4 (Urgency bound)**.** *Under the hypotheses of Theorem 3.3, $\Phi(d, R) \leq BR/(R+1) < B$.*

Proof. By Theorem 3.3, $|d| \leq BR$. Hence $\Phi(d, R) = |d|/(R+1) \leq BR/(R+1)$. Since $R/(R+1) < 1$ for all $R \geq 0$, we get $\Phi(d, R) < B$. □

Remark 3.5 (Interpretation). The urgency bound says the average debt per remaining decision is strictly less than the per-tick capacity. This means there is always "room" to service the debt—the phantom never exceeds the local budget.

4. Schedule neutrality and phase-correcting rotation

We now extend from a single window to an infinite schedule on $\mathbb{Z}$.

Definition 4.1 (Aligned W-tick schedule)**.** An integer signal $s : \mathbb{Z} \to \mathbb{Z}$ satisfies *W-tick neutrality* if $\sum_{k=0}^{W-1} s(t_0 + k) = 0$ for every $t_0 \equiv 0 \pmod{W}$.

Definition 4.2 (Window index process)**.** Fix an initial phase $\iota_0 \in \{0, \ldots, W-1\}$. Define $\mathrm{idx} : \mathbb{Z}_{\geq 0} \to \{0, \ldots, W-1\}$ by

$$\mathrm{idx}(0) := \iota_0, \qquad \mathrm{idx}(n+1) := (\mathrm{idx}(n) + 1) \bmod W.$$

This is a deterministic cyclic counter with period W.

Theorem 4.3 (Window index tracking)**.** *For the process in Definition 4.2,* $\mathrm{idx}(n) = (\iota_0 + n) \bmod W$ *for every* $n \geq 0$.

Proof. By induction on n. *Base case*: $\mathrm{idx}(0) = \iota_0 = (\iota_0 + 0) \bmod W$. ✓

Inductive step: assume $\mathrm{idx}(n) = (\iota_0 + n) \bmod W$. Then by Definition 4.2:

$$\mathrm{idx}(n+1) = (\mathrm{idx}(n) + 1) \bmod W = ((\iota_0 + n) \bmod W + 1) \bmod W = (\iota_0 + n + 1) \bmod W. \qquad \square$$

Definition 4.4 (Boundary-balance rule)**.** An integer signal $s : \mathbb{Z} \to \mathbb{Z}$ satisfies the *boundary-balance rule* if for every block index $j \geq 0$:

$$s(Wj + W - 1) = -\sum_{k=0}^{W-2} s(Wj + k).$$

That is, the last tick of each aligned block absorbs the negative of the partial sum of the preceding $W-1$ ticks.

Theorem 4.5 (Neutrality from boundary balance)**.** *If s satisfies the boundary-balance rule, then s satisfies W-tick neutrality: for every $j \geq 0$,*

$$\sum_{k=0}^{W-1} s(Wj + k) = 0.$$

Proof. Fix $j \geq 0$. Split the block sum:

$$\sum_{k=0}^{W-1} s(Wj + k) = \underbrace{\sum_{k=0}^{W-2} s(Wj + k)}_{=:P} + s(Wj + W - 1).$$

By the boundary-balance rule, $s(Wj + W - 1) = -P$. Hence the total is $P + (-P) = 0$. $\square$

Theorem 4.6 (Schedule neutrality rotation)**.** *Let s satisfy W-tick neutrality. Then there exists a unique offset $r \in \{0, \ldots, W-1\}$ such that for every $k \geq 0$:*

$$\sum_{j=0}^{W-1} s(r + Wk + j) = 0 \qquad \textit{and} \qquad \mathrm{idx}(r + Wk) = 0.$$

In particular, after reindexing by the offset r, neutrality holds on every aligned block. The offset is $r = (W - \iota_0) \bmod W$.

Proof. Set $r := (W - \iota_0) \bmod W$. By Theorem 4.3, $\mathrm{idx}(r) = (\iota_0 + r) \bmod W = (\iota_0 + W - \iota_0) \bmod W = 0$. Since $\mathrm{idx}(r) = 0$, the W consecutive ticks $r, r+1, \ldots, r+W-1$ form a complete aligned block starting at a multiple of W in the reindexed coordinate. By W-tick neutrality, their sum is zero. Repeating with $r + Wk$ for every $k \geq 0$ gives the full statement.

Uniqueness: if r' also satisfies $\mathrm{idx}(r') = 0$, then $r' \equiv r \pmod{W}$, so $r' = r$ in $\{0, \ldots, W-1\}$. $\square$

5. Phantom cost augmentation

When the window neutrality constraint is composed with a cost functional (such as the J-cost of [1]), the phantom balance acts as an additive penalty that lifts the effective cost.

Definition 5.1 (Augmented cost)**.** For a configuration $x > 0$, phantom data (d, R), and penalty scale $\lambda \geq 0$:

$$J_{\mathrm{PL}}(x; d, R) := J(x) + \lambda \cdot \Phi(d, R).$$

Proposition 5.2 (Augmented cost dominates base cost)**.** *$J_{\mathrm{PL}}(x; d, R) \geq J(x)$ for all $x, d, R, \lambda \geq 0$. Equality holds iff $d = 0$ or $\lambda = 0$.*

Proof. $\Phi(d, R) = |d|/(R + 1) \geq 0$ and $\lambda \geq 0$, so $\lambda\Phi \geq 0$. If $d \neq 0$ and $\lambda > 0$, then $\lambda\Phi > 0$ and the inequality is strict. □

Remark 5.3 (Phase-winding obstruction)*.* In applications to arithmetic phase dynamics (see [2]), the phantom mechanism forbids sustained one-sided phase winding: any phase debt accumulated at tick k must be balanced by tick $k+W-1$. The backward projection bound (Theorem 3.3) quantifies the maximum sustainable debt at each intermediate tick. Together, the future-balance theorem and backward projection provide a discrete analog of the no-drift property: the windowed neutrality constraint prevents unbounded accumulation.

Acknowledgments

The author thanks the referees for careful reading and for suggesting the generalization to arbitrary window width W.

References

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